

Jonathan Muhire

Software Engineer - Robotics Simulation, CI Reliability, ML Evaluation
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SUMMARY

Software engineer and M.S. AI candidate focused on C++/Python systems, robotics simulation, CI reliability, and ML evaluation. Recently submitted an upstream comma.ai openpilot PR for MetaDrive CI with simulator artifacts, observed-failure fixes, green upstream checks, and 20 consecutive full workflow passes on one submitted commit.

EDUCATION

Oklahoma Christian University

B.S. Computer Science and M.S. Artificial Intelligence - Concurrent Degrees

GPA: 3.89/4.0. Completing remaining requirements asynchronously; available for immediate full-time employment.

Expected Aug. 2026

Edmond, OK

ENGINEERING EXPERIENCE

comma.ai openpilot

Open Source Contributor - MetaDrive CI Bounty #30693

Jun. 2026

Remote

- Submitted PR #38242 adding Linux GitHub Actions MetaDrive simulator coverage, scoped dependency install, and simulator log artifact upload.
- Patched only observed failures from Actions logs and artifacts: lane/crash behavior, modeld restart handling, traffic-free vision leads, movement accounting, upload timing, racy Params timing, and runner apt-source 403s.
- Validated the branch with 20/20 consecutive full `tests.yaml` workflow executions, green upstream PR checks, and mergeable current head.

Neotix Robotics

Technical Lead and Founder

May 2025 to Present

New Haven, CT

- Received \$20,000 in non-dilutive support from Yale Tsai City Ventures to fund R&D on robotics data collection tooling.
- Built a pipeline to store and version roughly 500GB of RGB-D and geometry logs using MinIO and LakeFS, enabling reproducible dataset snapshots and rollbacks.
- Developed a ROS2/C++ teleoperation and data-capture prototype; integrated SAM2 segmentation and ORB-SLAM3 tracking into the perception stack.

Scale AI

AI Safety and Code Evaluation Specialist

May 2024 to Aug. 2024

San Francisco, CA

- Created adversarial and edge-case evaluations for code-generation models, focusing on security-sensitive prompts, robustness failures, async workflows, and API integrations.
- Reviewed and annotated 3,000+ lines of model-generated code, documenting injection risks, race conditions, memory bugs, and scoring rubrics.

RESEARCH AND OPEN SOURCE

RenAIssance - Document Understanding Pipeline

Google Summer of Code 2025 - Deep Learning Engineer

May 2025 to Aug. 2025

Remote

- Contributed to an open-source manuscript digitization pipeline using LayoutLMv3; implemented training, inference, evaluation, structured logging, input validation, and batch failure handling.

Data and ML Systems Research

Graduate Researcher

Jan. 2024 to Present

Edmond, OK

- Built Jerry Mandarin, a six-state election model over 1.89M census blocks; converted RDH/VEST shapefiles to CSVs and trained a k=512 MLP for R/D vote-share prediction (avg. MAE 0.1005).

SELECTED PROJECTS

Agent Evaluation - Memory Ablations

ReAct agent evaluation harness with memory-mode ablations and needle-in-haystack tests; eval-first development for retrieval and tool-use failure modes.

2025

Vesuvius Challenge - 3D Segmentation

Volumetric deep learning for CT scans of carbonized Herculaneum scrolls; focused on data handling, segmentation quality, and model evaluation.

2025

TECHNICAL SKILLS

Languages: C++, Python, SQL, TypeScript, JavaScript, MATLAB

Systems and Robotics: ROS2, GitHub Actions, Docker, Linux, CI/CD, telemetry, simulator debugging, data capture

ML and Vision: PyTorch, TensorFlow, Transformers, OpenCV, SAM2, ORB-SLAM3, LayoutLMv3, OCR

Data Infrastructure: MinIO, LakeFS, PostgreSQL, S3-compatible storage, ETL, reproducible datasets